Dylan Winters

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EDUCATION

MEng Electrical and Electronic Engineering – First Class Honours, Dean's List
Imperial College London | Oct 2022 – Jun 2026

- Coursework: Robotic Manipulation, Human-Centred Robotics, Optimisation, Deep Learning, Machine Learning, Control of Multi-Body Mechanical Systems

SELECTED PROJECTS

Compute Budgets for Humanoid Locomotion Policies
Self-initiated | Jun 2026 – Present

- Training PPO locomotion policies for the Unitree G1 in IsaacLab, sweeping control frequency to locate minimum compute budget before motion quality degrades
- Evaluating motion quality using velocity tracking, push-recovery robustness and RMS action jerk across multiple random seeds per configuration
- Extends a thesis finding that control frequency was a primary limiting factor in motion quality, and thus perceived system reliability, on embedded hardware

RIGBY – Low-Cost Autonomous Mobile Bimanual Manipulator
Master's Thesis, supervised by Prof. Demiris, Personal Robotics Lab | Sep 2025 – Jun 2026

- Built a £1,951 mobile bimanual robot end-to-end, spanning mechanical design, actuation, sensing, teleoperation, data collection and onboard policy deployment
- Designed a 7-DOF dominant arm and parallel-jaw grippers, optimized for joint loading while preserving workspace and grasp reliability
- Built the CAD, MJCF and URDF models for simulation; authored the full robot description and released it to facilitate further research and learning
- Developed whole-body teleoperation using leader arms and rudder pedals, collecting arm and base demonstrations simultaneously through LeRobot
- Proposed DOM-ACT, an extension of Action Chunking with Transformers adding an explicit dominant limb-support limbs structure; milestone and full autonomous completion success rates rose across all evaluated sequential bimanual tasks

TFLBot – Assistive Guidance Robot for Underground Networks
Human-Centred Robotics, team of 10 | Jan 2026 – Mar 2026

- Extended a SoftBank Pepper with a Jetson Orin Nano, RGB-D camera, microphone array and RPLIDAR over ROS 2 Humble
- Built a deictic gesture library through kinesthetic teaching, timed to align with speech output
- Used continuous evaluation to handle edge cases to improve system robustness

EXPERIENCE

Investment Analyst
Zuri-Invest | Aug 2026 – Present

- Building internal Python tooling to automate investment logging and research workflows across equities and structured products

Data Intern
Pactio Technologies Limited | Sep 2024 – Oct 2024

- Analysed 50 corporate acquisition proposals and structured the resulting dataset used to train and evaluate a machine learning model

Software Engineering Intern
Gaman Succession | Jun 2024 – Sep 2024

- Developed internal Python and SQL tools to automate investment research pipelines, processing 1m+ records daily and reducing time to generate fully qualified leads from 5 analyst-days to 1 analyst-hour

SKILLS

- Simulation and Modelling: MuJoCo, Isaac Lab, Isaac Sim, URDF, ROS 2, Fusion 360, 3D printing
- Machine Learning and Control: PyTorch, PPO, imitation learning (ACT, VLA), LeRobot, teleoperation, SLAM
- Programming: Python, MATLAB, SQL, Git, Linux
- Languages: English (native), French (native), German (C1)

ADDITIONAL INFORMATION

- Nationalities: Swiss, Irish and British; full right to work in the EU and UK
- Activities and interests: Imperial College Futsal and electronic music production